

Vathala Manikantha Narapa Reddy

Software Engineer | Full Stack | Distributed Systems

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🌐 LinkedIn 🐙 GitHub 🌐 Portfolio

Education

National Institute of Technology Hamirpur
B.Tech in Electronics and Communication Engineering

2019 – 2023
CGPA: 7.95

Professional Experience

Founder & Full Stack Engineer
Kara HMS

Apr 2024 – Present

- Founded and built a Hospital Management System as a solo founder.
- Architected a modular backend using NestJS with a loosely coupled persistence layer enabling migration from Google Sheets to a database with minimal changes.
- Built administrative dashboards, account management and the Outpatient (OP) workflow.
- Designed REST APIs, authentication and reusable backend modules for future IP, Pharmacy and Billing features.
- Led architecture, frontend, backend and deployment.

Software Development Engineer
Jio Platforms

Jul 2023 – Mar 2024

- Developed production REST APIs using Node.js and TypeScript in a microservice architecture.
- Built React-based enterprise applications and internal business tools.
- Implemented a role-based recommendation portal integrating ML-backed services.
- Developed FYND Platform extensions and resolved production issues.
- Received the **Best Debutant Award**.

Software Engineering Intern
SmartKnower

Dec 2021 – Jan 2022

- Added features to a MERN e-commerce application and improved frontend and backend quality.

Projects

PerpX – Perpetual Futures Exchange

Next.js, TypeScript, Redis, WebSockets

- Built a perpetual futures exchange with a low-latency matching engine.
- Designed an optimized order book with snapshot and recovery.
- Implemented isolated margin, liquidation engine, funding calculations and real-time market updates.
- Built event-driven services using Redis and WebSockets.
- Developed a responsive trading UI using Next.js.

AI Agent CLI

TypeScript, AI SDK

- Built an AI-powered CLI with structured tool calling using JSON Schema.
- Designed long-context conversation summarization for developer workflows.

Technical Skills

Languages	C++, TypeScript, JavaScript, Python, SQL
Frontend	React, Next.js, Tailwind CSS, HTML, CSS
Backend	Node.js, NestJS, Express.js, REST APIs, WebSockets
Databases	PostgreSQL, MongoDB, SQLite, Google Sheets API
Distributed Systems	Redis, Kafka, RabbitMQ
AI	Vercel AI SDK, Tool Calling, Streaming LLMs
Tools	Docker, Git, GitHub, Linux, Bun, AWS S3

Achievements

- Best Debutant Award – Jio Platforms.
- Codeforces Rating: 970.
- Certified MERN Full Stack Developer (100xDevs).